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Markov Chain Monte Carlo Studies of Ising Criticality in Two and Three Dimensions

Background

In two or more dimensions the zero field Ising model has a continuous phase transition at the critical temperature T_c: above T_c the system is disordered with spin flip symmetry, whereas below T_c this symmetry is spontaneously broken producing a net magnetisation. Onsager's work in 1944 established this critical behaviour in two dimensions, deriving a critical temperature of $T_c \approx 2.269 J/k_B$[1]. No exact solution is known for the 3D Ising model motivating high precision calculations with Monte Carlo methods finding $T_c = 4.51152326(1) J/k_B$ [2]. In the critical region close to T_c physical quantities have a universality following power laws characterised by critical exponents of t the reduced the temperature $t = (T - T_c)/T_c$ (1.2), regardless of the microscopic details of the system itself only depending on the system's dimension d and the order parameter symmetry. The 2D Ising model's critical exponents are known exactly: [3]

$$\alpha = 0, \qquad \beta = \frac{1}{8}, \qquad \gamma = \frac{7}{4}, \qquad \nu = 1, \tag{1.1}$$

$$c_v \sim |t|^{-\alpha}, \quad m \sim (-t)^{\beta}, \quad \chi \sim |t|^{-\gamma}, \quad \xi \sim |t|^{-\nu}. \tag{1.2}$$

Monte Carlo methods are used for numerically measuring observables in systems with many degrees of freedom. Rather than directly computing the intractable partition function Z, a Markov chain converges to the target Boltzmann distribution for sampling. In the Metropolis algorithm: a single spin flip is proposed and accepted probabilistically with $P = \min(1, e^{-\beta \Delta E})$. Metropolis is effective away from T_c though suffers from critical slowing down near criticality. At T_c the correlated spins grow to domain size ξ while Metropolis only flips one spin at a time leading to long autocorrelation times τ, leaving few independent samples for Monte Carlo estimation [4]. We can quantify this slowing down with the the dynamic exponent z, $\tau \sim |t|^{-\nu z} = \xi^z$. In a finite lattice at T_c the correlation length ξ is limited by the system size, so $\xi \to L$ and $\tau \sim L^z$ which can be measured. These can be calculated in 2D, for Metropolis $z \approx 2.17$ [5].

We introduce cluster flipping algorithms like Wolff [6] and Swendsen-Wang [7] to tackle critical slowing down, with both having $z = 0.25(1)$ in 2D [8]. These algorithms instead flip clusters of spins, growing clusters from aligned neighbours with probability $P_{\text{add}} = 1 - e^{-2\beta J}$. With these cluster algorithms, we can get improvements of τ a thousand times smaller than with Metropolis and we can get meaningful observables in the critical region for 100×100 lattices [5].

We can extract our critical exponents by looking at observables around T_c across various lattice sizes with the finite size scaling method. For example, the susceptibility peak is shifted away from T_c on a finite lattice, but sits at a fixed $x^\star$ of the scaling variable $x = tL^{\frac{1}{\nu}}$, independent of lattice size. Taking the susceptibility χ, we can find a maximum χ_{max} for various lattice sizes and recover $\frac{\gamma}{\nu}$. Similar methods can be applied to the other exponents.

$$\chi(t, L) = L^{\frac{\gamma}{\nu}} \underbrace{\chi_0(tL^{\frac{1}{\nu}})}_{\text{scaling fn}} = L^{\frac{\gamma}{\nu}} \chi_0(x), \qquad \chi_{\text{max}} = L^{\frac{\gamma}{\nu}} \chi_0(x^\star) = A L^{\frac{\gamma}{\nu}} \tag{1.3}$$

Goals

Starting with the 2D Ising framework we will compare Metropolis, Wolff and Swendsen-Wang critical slowing down effect at T_c and autocorrelation times at various lattice sizes, through measuring each algorithm's dynamic exponent z and cross checking autocorrelation methods, such as Sokal's windowing [4] and Wolff's automatic windowing (Γ-method) [9]. Using cluster algorithms for our simulations, we will extract critical exponents using finite-size scaling and estimate T_c using psuedo-critical temperatures and Binder cumulants. As the 2D case is known exactly we can verify our implementation by reproducing the known critical exponents from Baxter [3] and Onsager's T_c [1], both within the statistical error bars .

We aim to extend this computational framework to the three dimensional case, where two problems arise: finite size scaling is only exact for large L and higher order corrections to scaling are larger in 3D, as

$$T^\star(L) - T_c \sim L^{-\frac{1}{\nu}}(1 + A_1' L^{-\omega_i} + \cdots) \tag{1.4}$$

However scaling to larger L becomes more expensive as the cost per sweep grows with L^3 rather than L^2. As smaller lattice sizes are more strongly affected by higher order correction terms, we will check fits for stability with the lowest lattice size included, L_{min}. Finally comparing these results against modern high precision Monte Carlo work finding inverse critical temperatures and exponents, an example of which is $K_c = 0.221654626(5)$, $\nu = 0.629912(86)$ and $\gamma = 1.23708(33)$ all from Ferrenberg, Xu and Landau's 2018 work [2]. Our simulations will be benchmarked across Python and C++, scaling to the largest lattice size feasible with the use of the Seagull HPC, hopefully $L \ggg 100$. If time permits, we will explore Pseudo-Random Number Generators and their effect on error in measuring observables [10].

References

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